

NOTE: M appears because in FEM you are not working in ordinary $\mathbb{R}^n$ with the standard dot product; you are approximating the Hilbert space $L^2(M, d\text{vol}_g)$. The **mass matrix** M is exactly the finite-dimensional version of that inner product.

Continuous picture

In your notes, states live in

$$\mathcal{H} = L^2(M, d\text{vol}_g),$$

with inner product

$$\langle u, v \rangle = \int_{S^2} u(\bar{x}) v(x) d\text{vol}_g(x).$$

When you expand a wave function in FEM basis functions $\{\phi_i\}$,

$$u_h(x) = \sum_i u_i \phi_i(x),$$

the inner product becomes

$$\langle u_h, v_h \rangle = \sum_{i,j} \bar{u}_i v_j \int_M \phi_i \phi_j d\text{vol}_g.$$

The integrals $\int_M \phi_i \phi_j d\text{vol}_g$ are exactly the entries of the mass matrix:

$$M_{ij} = \int_M \phi_i(x) \phi_j(x) d\text{vol}_g(x).$$

So M is the matrix that tells you how to compute norms and inner products of discrete states in a way that still respects the manifold's metric.

Discrete eigenproblem

The continuous eigenproblem is

$$\hat{H}\psi = E\psi.$$

Projecting onto the FEM basis and using the L^2 inner product gives

$$\sum_j H_{ij} u_j = E \sum_j M_{ij} u_j,$$

so in matrix form:

$$Hu = EMu.$$

That's why a **generalized** eigenvalue problem appears. H is the matrix representation of your Hamiltonian in the $\{\phi_i\}$ basis, and M is the representation of the inner product. [file](#)

If you instead forced $M = I$ (standard dot product on coefficients), you would be implicitly changing the Hilbert space and losing the correct geometry and volume form $d\text{vol}_g$ that your notes spend time defining. [file](#)

Local FEM view

On each triangle T , for P1 elements,

$$M_T = \int_T \begin{pmatrix} \phi_1^2 & \phi_1\phi_2 & \phi_1\phi_3 \\ \phi_2\phi_1 & \phi_2^2 & \phi_2\phi_3 \\ \phi_3\phi_1 & \phi_3\phi_2 & \phi_3^2 \end{pmatrix} dS = \frac{|T|}{12} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

Assembling these local blocks over all triangles gives the global mass matrix M .

So M appears because:

- you expanded ψ in non-orthonormal FEM basis functions, and
- you want your discrete problem to live in the same geometric L^2 space your continuous theory is written in.

That's why the proper discrete Hamiltonian eigenproblem is $Hu = EMu$, not just $Hu = Eu$.

For our chosen potential we'll be taking into consideration the geodesic harmonic trap

A geodesic harmonic trap is just the curved-space version of the ordinary harmonic oscillator: instead of using Euclidean distance from a center, you use **geodesic distance along the surface**. So, the potential is

$$V(x) = \frac{k}{2} d_g(x, x_0)^2,$$

where x_0 is the trap center and d_g is the intrinsic distance on the manifold. This is the natural oscillator-style potential on a curved space and is commonly written in geodesic coordinates for constant-curvature spaces.

What each symbol means

- x : the point where you evaluate the potential.
- x_0 : the point on the surface where the trap is centered.

- $d_g(x, x_0)$: shortest distance **along the surface**, not through ambient $\mathbb{R}^3$.
- k : trap strength; bigger k means tighter confinement.

So near x_0 , the particle sees a bowl-shaped potential, just like the flat harmonic oscillator, except the distance is measured intrinsically on the manifold.

Sphere version

Since we're working on a sphere, the geodesic distance between two points x and x_0 on a sphere of radius R is the great-circle distance

$$d_g(x, x_0) = R \arccos \left(\frac{x \cdot x_0}{R^2} \right).$$

That is the standard spherical-distance formula.

So, our potential becomes

$$V(x) = \frac{k}{2} R^2 \arccos^2 \left(\frac{x \cdot x_0}{R^2} \right).$$

The source term **f**: definition and structure

We'll take **f** to be:

$$f(x) = A e^{i\ell\varphi(x)} \exp \left(-\frac{d_g(x, x_0)^2}{\sigma^2} \right)$$

$$x \in S^2 \subset \mathbb{R}^3.$$

$x_0 \in S^2$ is the center (you'll probably take x_0 as the north pole).

$d_g(x, x_0)$ is the geodesic distance on the sphere:

$$d_g(x, x_0) = \arccos(x \cdot x_0) \text{ on the unit sphere.}$$

$\varphi(x)$ is the azimuthal angle (longitude). In spherical coordinates (θ, φ) , this is your usual φ (adjust for your swapped θ, ϕ convention).

$A \in \mathbb{C}$ (or $\mathbb{R}$) is an amplitude.

$\ell \in \mathbb{Z}$ controls the angular momentum content.

$\sigma > 0$ sets the geodesic width of the Gaussian.

Decomposition into triangle contributions

Let the mesh be a triangulation $\mathcal{T}_h$ of (a discrete approximation of) S^2 . Then

$$F_i = \sum_{T \in \mathcal{T}_h} \int_T f(x) \phi_i(x) d\text{vol}_g(x),$$

and for each triangle T with vertices (x_a, x_b, x_c) you have local basis functions ϕ_a, ϕ_b, ϕ_c (the nodal hats restricted to T). doc.freefem+1

For a fixed triangle T , write the local load vector

$$F^T = \begin{pmatrix} F_a^T \\ F_b^T \\ F_c^T \end{pmatrix}, F_a^T = \int_T f \phi_a d\text{vol}_g, \text{ etc.}$$

Then

$$F_i = \sum_{T \ni i} F_i^T.$$

So the entire discrete RHS is built by looping over triangles and adding these local contributions to the global indices, exactly as you do for stiffness/mass.

On your sphere, you approximate $d\text{vol}_g$ by the Euclidean area of the embedded triangle so numerically your M_T has exactly this form with area as parameter.

Now,

$$F_p^T = \int_T f(x) \phi_p(x) d\text{vol}_g(x).$$

If f were constant on T , say $f(x) \approx f_T$ for all $x \in T$, then

$$F_p^T \approx f_T \int_T \phi_p(x) d\text{vol}_g(x).$$

But

$$\int_T \phi_p(x) d\text{vol}_g(x) = \sum_{q \in \{a,b,c\}} \int_T \phi_p(x) \phi_q(x) d\text{vol}_g(x) \cdot 1 = \sum_q (M_T)_{pq},$$

since $\phi_a + \phi_b + \phi_c \equiv 1$ on T .

Therefore

$$F_p^T \approx f_T \sum_q (M_T)_{pq},$$

or, vectorially,

$$F^T \approx M_T \mathbf{1} \quad f_T,$$

where $\mathbf{1} = (1,1,1)^T$.

This is exactly the pattern: the local load vector is the local mass matrix times a vector of ones, scaled by an appropriate sample of f on the triangle.

FINDING THE ANALYTIC SOLUTION USING GEOMETRIC FOURIER TRANSFORM

If you were to guess the Fourier transform derived from the geometry of the sphere, you'd say something like:

$$\mathcal{F}[f](\omega) = \int_{S^2} f(x) e^{-2\pi i \langle x, \omega \rangle_g} d\text{vol}_g$$

The problem with this formula is that it's not exactly why the Fourier transform works on PDEs in the first place.

The notion of defining the usual Fourier transform was that $\hat{f}(\omega)$ was the projection of f onto the eigen basis of $\frac{d}{dx}$, as a linear operator. The thing is that the exponential here is not an eigenfunction of our differential Δ_g

In standard spherical coordinates (θ, φ) , with ϕ = colatitude, θ = longitude,

$$\Delta_g u = \frac{1}{\sin^2 \phi} \frac{\partial^2 u}{\partial \theta^2} + \frac{1}{\sin \phi} \frac{\partial}{\partial \phi} \left(\sin \phi \frac{\partial u}{\partial \phi} \right)$$

look for eigenfunctions

$$-\Delta_g Y(\theta, \varphi) = \lambda Y(\theta, \varphi).$$

Try separated variables: $Y(\theta, \varphi) = \Theta(\theta)\Phi(\varphi)$. Plugging in and separating gives two ODEs:

Azimuthal Equation:

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\varphi^2} = -m^2 \Rightarrow \Phi(\varphi) = e^{im\varphi}$$

Polar equation (In $x = \cos(\theta)$):

$$(1-x^2)\frac{d^2\theta}{dx^2} - 2x\frac{d\theta}{dx} + (\ell(\ell+1) - \frac{m^2}{1-x^2})\theta = 0$$

whose solutions are the **associated Legendre functions** $P_\ell^m(x)$

Putting the pieces together,

$$Y_\ell^m(\theta, \varphi) = N_{\ell m} P_\ell^m(\cos \theta) e^{im\varphi},$$

with a normalization constant chosen so that

$$\int_{S^2} Y_\ell^m \overline{Y_{\ell'}^{m'}} d\Omega = \delta_{\ell\ell'} \delta_{mm'}.$$

A standard choice is

$$Y_\ell^m(\theta, \varphi) = \sqrt{\frac{(2\ell+1)(\ell-m)!}{4\pi(\ell+m)!}} P_\ell^m(\cos \theta) e^{im\varphi}.$$

So now we project onto the eigen-basis of our operator to get

$$\mathcal{F}[f](\ell, m) = \int_{S^2} f(\theta, \varphi) \overline{Y_\ell^m(\theta, \varphi)} d\text{vol}_g,$$

Is $\mathcal{F}$ well-defined?

In other words, if we define the map

$$\begin{aligned} \mathcal{F}: L^2(S^2) &\rightarrow \ell^2(\{(\ell, m)\}) \\ f &\rightarrow (\hat{f}_{\ell m}) = \left(\int_{S^2} f(\theta, \varphi) Y_\ell^m(\theta, \varphi) d\text{vol}_g \right) \end{aligned}$$

Do we get a well-defined map?

And one can see using basic integration theory and Plancherel that we do have a well-defined map as well as

$$\|f\|_{L^2}^2 = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} |\hat{f}_{\ell m}|^2$$

For $s \geq 0$, the Sobolev space $H^s(S^2)$ can be described via the spherical harmonic expansion. A standard equivalent definition is:

- $f \in H^s(S^2)$ iff $f \in L^2(S^2)$ and

$$\|f\|_{H^s(S^2)}^2 \sim \sum_{\ell=0}^{\infty} (1 + \ell(\ell+1))^s S_{ff}(\ell),$$

where $S_{ff}(\ell) = \sum_{m=-\ell}^{\ell} |\hat{f}_{\ell m}|^2$ is the power spectrum at degree ℓ .

In other words, the Sobolev norm of order s is equivalent to a weighted ℓ^2 -norm of the spherical harmonic coefficients, with weights $(1 + \lambda_{\ell})^s$ where $\lambda_{\ell} = \ell(\ell+1)$ are the Laplace–Beltrami eigenvalues.

This shows explicitly:

$\mathcal{F}$ is a linear isomorphism between $H^s(S^2)$ and the sequence space

$$\left\{ \hat{f}_{\ell m} : \sum_{\ell, m} (1 + \ell(\ell+1))^s |\hat{f}_{\ell m}|^2 < \infty \right\}.$$

The norm on $H^s(S^2)$ is equivalent to the norm on this coefficient space.

So let's apply the transform to our strong solution

The strong-form equation is

$$-\frac{\hbar^2}{2m} \Delta_g u + V u = f.$$

Apply the spherical harmonic transform: take inner products with $Y_{\ell}^{\bar{m}}$ on both sides. Using orthonormality, you get for each (ℓ, m) :

$$\frac{\hbar^2}{2m} \lambda_{\ell} u_{\ell m} + \int_{S^2} V u Y_{\ell}^{\bar{m}} d\text{vol}_g = f_{\ell m}.$$

If V is **constant** (say $V \equiv V_0$), then

$$\int_{S^2} V_0 u Y_{\ell}^{\bar{m}} d\text{vol}_g = V_0 u_{\ell m},$$

and the mode equation simplifies to

$$\left(\frac{\hbar^2}{2m} \lambda_{\ell} + V_0 \right) u_{\ell m} = f_{\ell m}.$$

Solving one would get:

$$u_{\ell m} = \frac{f_{\ell m}}{\frac{\hbar^2}{2m} \ell(\ell+1) + V_0}.$$

So our solution would yield

$$u(\theta, \varphi) = \sum_{\ell, m} u_{\ell m} Y_m^\ell(\theta, \varphi)$$

to test the FEM method more concretely we'll substitute our V , with a particular V_0 ...

There are two natural ways to compare:

A. Compare in physical space (nodal values)

1. Choose a test mesh refinement level and fix all parameters (A_0, ℓ, σ, V_0) . [perplexity](#)
2. For each vertex x_i of your mesh, evaluate the analytic solution

$$u_{\text{analytic}}(x_i) = \sum_{\ell=0}^{L_{\max}} \sum_{m=-\ell}^{\ell} u_{\ell m} Y_\ell^m(\theta, \varphi)$$

truncated at some $L_{\max}$ based on your resolution.

3. Compare the FEM nodal values $u_h(x_i) = u_i$ to $u_{\text{analytic}}(x_i)$, for example via

$$\max_i |u_i - u_{\text{analytic}}(x_i)| = \left(\sum_i |u_i - u_{\text{analytic}}(x_i)|^2 \right)^{1/2}.$$

4. Refine the mesh (increase subdivisions) and check that this error decays roughly like some power of the mesh size h (first order for P1 elements). [people.tamu.edu](#)

This is the most direct “are my numbers right?” comparison.

B. Compare in spectral space (project FEM onto spherical harmonics)

1. Interpret u_h as a function on the sphere (piecewise linear).
2. Approximate its spherical harmonic coefficients numerically:

$$\hat{u}_{\ell m}^{(h)} \approx \int_{S^2} u_h(\theta, \varphi) Y_\ell^m(\bar{\theta}, \varphi) d\text{vol}_g,$$

using a quadrature on the sphere (or sampling at mesh vertices with appropriate weights).

3. Compare $\hat{u}_{\ell m}^{(h)}$ to the analytic $u_{\ell m}$:

$$|\hat{u}_{\ell m}^{(h)} - u_{\ell m}|$$

as ℓ varies.

4. Again, refine the mesh and see the spectral error shrink.

This approach is conceptually neat because it uses the same basis (spherical harmonics) in theory and in your comparison.

To begin implementing the error analysis code (using method B) we'll start by discussing

$$f_{\ell m} = \sqrt{\frac{(2\ell+1)(\ell-m)!}{4\pi(\ell+m)!}} \int_0^{2\pi} \int_0^\pi f(\theta, \varphi) P_\ell^m(\cos \theta) e^{-im\varphi} \sin \theta d\theta d\varphi.$$

But For our specific source

$$f(\theta, \varphi) = A e^{i\ell_0 \varphi} \exp\left(-\frac{d_g((\theta, \varphi), x_0)^2}{\sigma^2}\right),$$

the spherical harmonic coefficient integral becomes explicit but not closed-form elementary. Let's write it carefully and see what structure we get.

Then

$$f_{\ell m} = A N_{\ell m} \int_0^{2\pi} \int_0^\pi e^{i\ell_0 \varphi} e^{-\theta^2/\sigma^2} P_\ell^m(\cos \theta) e^{-im\varphi} \sin \theta d\theta d\varphi,$$

$$\text{where } N_{\ell m} = \sqrt{\frac{(2\ell+1)(\ell-m)!}{4\pi(\ell+m)!}}$$

Factor the φ -dependence:

$$f_{\ell m} = A N_{\ell m} \left(\int_0^{2\pi} e^{i(\ell_0-m)\varphi} d\varphi \right) \left(\int_0^\pi e^{-\theta^2/\sigma^2} P_\ell^m(\cos \theta) \sin \theta d\theta \right).$$

The φ -integral: selection of $m = \ell_0$

Compute

$$\int_0^{2\pi} e^{i(\ell_0-m)\varphi} d\varphi = \begin{cases} 2\pi, & m = \ell_0, \\ 0, & m \neq \ell_0. \end{cases}$$

So you get **exactly one azimuthal index**:

$$f_{\ell m} = 0 \text{ if } m \neq \ell_0,$$

and for $m = \ell_0$,

$$f_{\ell,\ell_0} = 2\pi A N_{\ell,\ell_0} \int_0^\pi e^{-\theta^2/\sigma^2} P_\ell^{\ell_0}(\cos \theta) \sin \theta d\theta.$$

So it only matters for us to solve $u_{\ell m}$ when $m = \ell_0$

$$u_{\ell\ell_0} = \frac{f_{\ell\ell_0}}{\frac{\hbar^2}{2m} \ell(\ell+1) + V_0}$$

And therefore

$$u = \sum_{\ell=0}^{\infty} u_{\ell\ell_0} Y_\ell^{\ell_0}(\theta, \varphi)$$

To compare the error of our calculated FEM we decided to switch to a constant potential and then find the L^2 difference between u_h and u . one must be vigilant with the use of the norm, since we're using the mesh's geometry, we compute our global mass matrix and use it as our metric.

So first

$$err_h = u_h - u$$

And then

$$\|err_h\|_{L^2} = \|err_h^\dagger \cdot (Merr_h)\|_2$$

Now we are mostly comfortable with the idea of a metric and manifold and the manipulation of local coordinates and the idea of solving a PDE on a given manifold, let's turn our attention towards solving for equations involving differential forms on a manifold.

Let M^n be a Riemann Manifold and g it's metric

HODGE STAR $\star$

The Hodge star $\star$ is the operator that takes a k -form to an $(n - k)$ -form using the metric and orientation

It depends on both the metric and the orientation.

Inner product on covectors and forms

- At each p , g_p gives an inner product on $T_p M$ and hence on $T_p^* M$. [dec41.srcf](#)

- Extend this to $\Lambda^k T_p^* M$: if

$$\alpha = v_1^b \wedge \cdots \wedge v_k^b, \beta = w_1^b \wedge \cdots \wedge w_k^b$$

(with $\flat$ the musical isomorphism $v \mapsto g(v, \cdot)$), define

$$\langle \alpha, \beta \rangle = \det(g(v_i, w_j)),$$

and extend bilinearly.

is the **dual** of $T_p M$: its elements are linear maps $T_p M \rightarrow \mathbb{R}$. Since g_p is non-degenerate, there is a canonical isomorphism (the **flat/sharp** map):

$$\flat: T_p M \xrightarrow{\sim} T_p^* M, v \mapsto v^\flat := g_p(v, \cdot).$$

Its inverse is $\sharp = \flat^{-1}$

This lets you **transfer** the inner product from $T_p M$ to $T_p^* M$: given two covectors $\alpha, \beta \in T_p^* M$, define

$$\langle \alpha, \beta \rangle_{T_p^* M} = g_p(\alpha^\sharp, \beta^\sharp).$$

In coordinates, if (g^{ij}) is the **inverse metric matrix**, then

$$\langle dx^i, dx^j \rangle_{T_p^* M} = g^{ij}(p).$$

So the inner product on covectors is computed using g^{-1} , not g itself.

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A k -form at p is an element of $\Lambda^k T_p^* M$: an **antisymmetric** multilinear functional on k tangent vectors. The inner product on $\Lambda^k T_p^* M$ is induced from $T_p^* M$ via the following rule.

For **decomposable** (pure wedge) k -forms:

$$\alpha = v_1^b \wedge \cdots \wedge v_k^b, \quad \beta = w_1^b \wedge \cdots \wedge w_k^b,$$

define

$$\langle \alpha, \beta \rangle_{\Lambda^k T_p^* M} = \det(g_p(v_i, w_j))_{i,j=1}^k$$

and **extend bilinearly** to all $\Lambda^k T_p^* M$.

L^2 -inner product on forms

- For $\alpha, \beta \in \Omega^k(M)$, their **pointwise inner product** is the function

$$\langle \alpha, \beta \rangle(p) = \langle \alpha_p, \beta_p \rangle_{\Lambda^k T_p^* M}$$

defined as above

- The global L^2 inner product is

$$(\alpha, \beta)_{L^2} = \int_M \langle \alpha, \beta \rangle \text{vol}_g.$$

- This is what δ is adjoint to, and what Hodge theory uses.

Definition: At each point $p \in M$, $\star_p: \Lambda^k T_p^* M \rightarrow \Lambda^{n-k} T_p^* M$ is the unique linear map such that

$$\forall u, v \in \Lambda^k T_p^* M: \quad u \wedge (\star v) = \langle u, v \rangle_{\Lambda^k} d\text{vol}_g$$

Basic properties (Riemannian case): wikipedia+2

- $\star$ is linear and an isomorphism: $\star \star \alpha = (-1)^{k(n-k)} \alpha$ for $\alpha \in \Omega^k(M)$.
- It depends on both the metric and the orientation; reversing the orientation flips the sign of $\star$ on odd degrees.
- Locally, in an oriented orthonormal coframe $\{e^1, \dots, e^n\}$,

$$\star(e^{i_1} \wedge \cdots \wedge e^{i_k}) = \epsilon e^{j_1} \wedge \cdots \wedge e^{j_{n-k}},$$

where $(i_1, \dots, i_k, j_1, \dots, j_{n-k})$ is a permutation of $(1, \dots, n)$ and $\epsilon = \pm 1$ its sign.

The Hodge star realizes an isometry (up to sign) between Ω^k and Ω^{n-k} :

$$\langle \alpha, \beta \rangle \text{dvol}_g = \alpha \wedge \star \beta.$$

This identity is the key to relating d and δ .

The Hodge star is the metric-dependent isomorphism that identifies k -forms with $(n-k)$ -forms by declaring that “wedge with $\star \beta$ ” realizes the inner product with β , times the volume form. Geometrically, it turns oriented k -dimensional volume elements into their complementary $(n-k)$ -dimensional volume elements, with signs and factors dictated by the metric (Euclidean or Lorentzian).

In 3D: area bivector $\leftrightarrow$ normal vector

In oriented Euclidean 3-space, an oriented plane with an area element can be represented by a bivector (or a 2-form) like $u \wedge v$. The Hodge star sends this bivector to the unique normal vector whose length equals the area of that parallelogram, with the orientation chosen consistently.

Concretely:

- Take two vectors $u, v \in \mathbb{R}^3$. Their wedge $u \wedge v$ is a bivector encoding the oriented area. [alexkritchevsky+1](#)
- The Hodge star satisfies

$$\star (u \wedge v) = u \times v,$$

i.e. it turns the area bivector into the usual cross-product normal vector. [wikipedia+1](#)

- Conversely, a normal vector n to a plane corresponds (via $\star$) to the oriented area bivector of the plane perpendicular to n .

A worked Hodge–Laplace calculation on a scalar field

Let’s make one explicit example in flat Minkowski to see the role of $\star$.

Take $\phi(t, x, y, z) = f(t) g(x, y, z)$ a separable scalar field, say with g rotationally symmetric in space. Then:

1. Compute $d\phi$:

$$d\phi = \partial_t \phi dt + \partial_x \phi dx + \partial_y \phi dy + \partial_z \phi dz.$$

2. Compute $\star d\phi$ using the Minkowski $\star$ rules on 1-forms (similar to the 2-form case, but now mapping to 3-forms):

$$\star(dt) = dx \wedge dy \wedge dz, \star(dx) = dt \wedge dy \wedge dz, \text{etc.}$$

So

$$\star d\phi = (\partial_t \phi) dx \wedge dy \wedge dz + (\partial_x \phi) dt \wedge dy \wedge dz + \dots$$

3. Compute $d(\star d\phi)$; you get a 4-form whose coefficients are combinations of second derivatives of ϕ .
4. Apply $\star$ again to turn that 4-form back into a scalar (0-form). Up to sign conventions, you obtain ϕ .

The **codifferential** δ is the formal adjoint of the exterior derivative d with respect to the L^2 inner product.

By definition:

$$\text{For compactly supported forms, } (d\alpha, \beta)_{L^2} = (\alpha, \delta\beta)_{L^2}$$

Using the Hodge star identity, one can derive an explicit formula: On an oriented Riemannian n -manifold,

$$\delta = (-1)^{n(k+1)+1} \star d \star$$

acting on k -forms. (Different sign conventions exist; the essential point is that δ is $\star d \star$ up to a sign depending on n and k .)

- It decreases degree by 1: $\delta: \Omega^{k+1} \rightarrow \Omega^k$.
- It satisfies $\delta^2 = 0$ just like $d^2 = 0$.

Intuitively, δ is the “divergence” operator on forms, generalizing the divergence of a vector field, while d is the “curl/gradient” side.

In local coordinates with metric g , δ can be expressed in terms of covariant derivatives and the Levi-Civita connection, e.g. for 1-forms α ,

$$(\delta\alpha) = -\nabla^i \alpha_i,$$

where indices are raised with g^{ij} and ∇ is the Levi-Civita connection; this is the divergence. Take the unit 2-sphere $S^2 \subset \mathbb{R}^3$ with the standard round metric. In spherical coordinates (θ, φ) with $\theta \in (0, \pi)$ (colatitude) and $\varphi \in (0, 2\pi)$ (longitude), the metric is

$$g = d\theta^2 + \sin^2 \theta d\varphi^2.$$

From this:

- Orthonormal coframe:

$$e^1 = d\theta, e^2 = \sin \theta d\varphi.$$

Then $g = (e^1)^2 + (e^2)^2$.

- Volume form:

$$\text{vol}_g = \sin \theta d\theta \wedge d\varphi = e^1 \wedge e^2.$$

In this orthonormal frame, the Hodge star on simple forms is particularly nice:

$$\star 1 = e^1 \wedge e^2, \star e^1 = e^2, \star e^2 = -e^1, \star (e^1 \wedge e^2) = 1,$$

since $\star$ is defined by $u \wedge \star v = \langle u, v \rangle \text{vol}_g$ and $\{e^1, e^2\}$ is positively oriented orthonormal.

Here $n = 2$. For 1-forms ($k = 1$), the exponent is

$$n(k + 1) + 1 = 2 \cdot 2 + 1 = 5,$$

so

$$\delta = - \star d \star \text{ on 1-forms.}$$

Let's take a 1-form α on S^2 . To keep things concrete, choose

$$\alpha = f(\theta) d\theta,$$

where f is a smooth function of θ only (axisymmetric case). We'll compute $\delta\alpha$ in two ways.

Method 1: via $\delta = - \star d \star$

First rewrite α in the orthonormal coframe:

$$\alpha = f(\theta) d\theta = f(\theta) e^1.$$

1. Apply $\star$:

$$\star \alpha = \star (f e^1) = f \star e^1 = f e^2 = f(\theta) \sin \theta d\varphi.$$

Take the exterior derivative:

$$d(\star \alpha) = d(f(\theta) \sin \theta d\varphi).$$

Since the coefficient depends only on θ ,

$$d(\star \alpha) = \frac{d}{d\theta} (f(\theta) \sin \theta) d\theta \wedge d\varphi.$$

Finally apply $\delta = -\star d \star$:

$$\delta \alpha = -\star d(\star \alpha) = -\frac{1}{\sin \theta} \frac{d}{d\theta} (f(\theta) \sin \theta).$$

Thus, for our axisymmetric 1-form $\alpha = f(\theta) d\theta$,

$$\boxed{\delta \alpha(\theta, \varphi) = -\frac{1}{\sin \theta} \frac{d}{d\theta} (f(\theta) \sin \theta),}$$

The divergence of a vector field $X = X^\theta \partial_\theta + X^\varphi \partial_\varphi$ on S^2 is

$$\nabla_i X^i = \frac{1}{\sqrt{\det g}} \partial_i (\sqrt{\det g} X^i),$$

Now do the same computation from the coordinate formula

$$(\delta \alpha) = -\nabla_i \alpha^i,$$

with $\alpha = \alpha_\theta d\theta + \alpha_\varphi d\varphi$.

Our α has components

$$\alpha_\theta = f(\theta), \alpha_\varphi = 0.$$

Raise the index using the inverse metric g^{ij} :

$$g^{\theta\theta} = 1, g^{\varphi\varphi} = \frac{1}{\sin^2 \theta}, g^{\theta\varphi} = 0.$$

So

$$\alpha^\theta = g^{\theta\theta} \alpha_\theta = f(\theta), \alpha^\varphi = 0.$$

The divergence of a vector field $X = X^\theta \partial_\theta + X^\varphi \partial_\varphi$ on S^2 is

$$\nabla_i X^i = \frac{1}{\sqrt{\det g}} \partial_i (\sqrt{\det g} X^i),$$

with $\sqrt{\det g} = \sin \theta$.

Apply this to α^i :

$$\nabla_\theta \alpha^\theta + \nabla_\varphi \alpha^\varphi = \frac{1}{\sin \theta} \partial_\theta (\sin \theta \alpha^\theta) + \frac{1}{\sin \theta} \partial_\varphi (\sin \theta \alpha^\varphi).$$

Since $\alpha^\varphi = 0$ and $\alpha^\theta = f(\theta)$,

Thus

$$\delta \alpha = -\nabla_i \alpha^i = -\frac{1}{\sin \theta} \frac{d}{d\theta} (\sin \theta f(\theta)),$$

Hodge Laplacian Δ

The **Hodge Laplacian** (or Hodge–de Rham Laplacian) on forms is defined by

$$\Delta = d\delta + \delta d.$$

Key properties:

- $\Delta: \Omega^k(M) \rightarrow \Omega^k(M)$: it preserves degree.
- It is **self-adjoint** and **nonnegative** with respect to the L^2 inner product:

$$(\Delta \alpha, \alpha)_{L^2} = \|d\alpha\|^2 + \|\delta \alpha\|^2 \geq 0$$

for suitable α .

- On functions (0-forms), δd is the usual Laplace–Beltrami operator, so $\Delta f = \delta df = -\operatorname{div} \operatorname{grad} f$ (up to sign convention).
- A form α is **harmonic** if $\Delta \alpha = 0$. Hodge theory shows this is equivalent to $d\alpha = 0$ and $\delta \alpha = 0$ on compact manifolds.

Analytically, Δ is a second-order elliptic operator on a Riemannian manifold, generalizing the scalar Laplacian to higher-degree forms

On a compact oriented Riemannian manifold, smooth forms decompose orthogonally as

$$\Omega^k(M) = \mathcal{H}^k \oplus d\Omega^{k-1}(M) \oplus \delta\Omega^{k+1}(M)$$

where $\mathcal{H}^k = \ker \Delta$ is the finite-dimensional space of harmonic k -forms. This is the Hodge decomposition theorem in the form most useful for PDE and geometry.

The statement is deeper than it first appears. It says every smooth form splits uniquely into:

- a harmonic part, carrying topology;
- an exact part, built from d ;

- a coexact part, built from δ .

So Hodge decomposition is not just a decomposition of functions or fields; it is the canonical decomposition into topology plus two mutually orthogonal differential pieces.

A chain complex $(C_\bullet, d_\bullet)$ over a ring is a sequence of modules $\{C_n\}_{n \in \mathbb{Z}}$ with maps $d_n: C_n \rightarrow C_{n-1}$ such that $d_{n-1} \circ d_n = 0 \forall n$

A co-chain complex $(C^\bullet, d^\bullet)$ is a sequence $\{C^n\}_{n \in \mathbb{Z}}$ with maps $d^n: C^n \rightarrow C^{n+1}$ such that $d^{n+1} \circ d^n = 0$. The only difference is whether degrees go up or down.

In both cases, $d^2 = 0$ means “image is contained in kernel”: for every k ,

$$\text{im } d_k \subset \ker d_{k-1}$$

or in the co-chain grading $\text{im } d^{k-1} \subset \ker d^k$.

For each degree k ,

- $Z^k = \ker d^k$: k -co-cycles
- $B^k = \text{im } d^{k-1}$: k -co-boundaries.
- $H^k = Z^k / B^k$: k -th co-homology group

Definition (cochain map). Given cochain complexes $(C^\bullet, d_C)$ and $(D^\bullet, d_D)$, a cochain map $f: C^\bullet \rightarrow D^\bullet$ is a collection of linear maps $f^k: C^k \rightarrow D^k$ such that

$$d_D^k \circ f^k = f^{k+1} \circ d_C^k$$

Because f sends cycles to cycles and boundaries to boundaries, it induces a well-defined linear map

$$H^k(f): H^k(C^\bullet) \rightarrow H^k(D^\bullet)$$

by sending a class $[\alpha]$ to $[f^k(\alpha)]$.

Definition (cochain homotopy). Two cochain maps $f, g: C^\bullet \rightarrow D^\bullet$ are cochain homotopic if there exists a collection of maps $h^k: C^k \rightarrow D^{k-1}$ such that

$$g^k - f^k = d_D^{k-1} \circ h^k + h^{k+1} \circ d_C^k$$

for all k

Theorem. If f and g are cochain homotopic, then $H^k(f) = H^k(g)$ for all k . In other words, cohomology is invariant under cochain homotopy.

For a smooth manifold M , the de Rham complex is the cochain complex

$$0 \rightarrow \Omega^0(M) \rightarrow \Omega^1(M) \rightarrow \Omega^2(M) \rightarrow \Omega^3(M) \rightarrow \dots$$

Where $\Omega^k(M)$ is the space of smooth differential k -forms and d is the exterior derivative. We have $d^2 = 0$, so exact forms (those of the form $d\beta$) are automatically closed (their derivative vanishes).

- $Z^k(M) = \{\alpha \in \Omega^k(M) \mid d\alpha = 0\}$: *closed k -forms*.
- $B^k(M) = \{\alpha \in \Omega^k(M) \mid \alpha = d\beta \text{ for some } \beta \in \Omega^{k-1}(M)\}$
- *The k -th de Rham cohomology group*

$$H_{dR}^k(M) = \frac{Z^k(M)}{B^k(M)}$$

a nonzero class $[\alpha] \in H_{dR}^k(M)$ is represented by a closed form α that cannot be written globally as $d\beta$, and such classes are in bijection with certain topological features of M (“holes,” non-trivial cycles).

Physical meaning: potentials, fields, and cohomology

In classical field theory and electromagnetism, you often have fields that are derivatives of potentials (e.g. $F = dA$), but global potentials exist only if certain cohomological obstructions vanish. [reddit+1](#)

- A curl-free field (closed 1-form) on a domain with trivial first cohomology admits a global potential; with non-trivial H^1 , there can be “cohomology pieces” not captured by any scalar potential. [ncatlab+1](#)
- More generally, the number of independent global constraints for the existence of electric or magnetic potentials is controlled by Betti numbers of the domain; de Rham cohomology thus generalizes familiar statements about Maxwell fields on multiply connected regions.

This same structure reappears in quantum mechanics when you look at gauge potentials, Berry connections, and supersymmetric quantum mechanics on manifolds.

One of the great consequences of Hodge theory is that each de Rham cohomology class has a unique harmonic representative on a compact oriented Riemannian manifold. Equivalently,

$$\mathcal{H}^k = H_{dR}^k(M)$$

This converts a topological object into the solution space of an elliptic PDE

In gauge theories and Maxwell-type equations, harmonic forms are the residual modes that cannot be removed by exact or co-exact adjustments alone. On compact spaces, they encode topological sectors, residual gauge freedoms, and obstructions to invertibility of the Laplacian unless one restricts to the orthogonal complement of harmonic forms.

Equivalently, every form ω can be uniquely written as

$$\omega = \underbrace{\gamma}_{\text{harmonic}} + \underbrace{d\alpha}_{\text{exact}} + \underbrace{\delta\beta}_{\text{coexact}},$$

with γ harmonic, $d\alpha$ in the d -image, and $\delta\beta$ in the δ -image, and all three pieces are mutually orthogonal in the L^2 inner product induced by the metric.

That is why solving $\Delta A = J$ is never only an operator inversion problem. One must ask whether orthogonal is harmonic forms, and whether a gauge or normalization condition has been imposed to remove the kernel.

In the spirit of scalar Sobolev spaces, one defines spaces in which the coefficients of the form and certain weak derivatives are square integrable. Etnyre's notes emphasize that Sobolev spaces of forms are completions of smooth forms in Sobolev norms and are the correct environment for elliptic theory on forms.

For first-order Hodge problems, natural space is often

$$H\Lambda^k(\Omega) = \{\omega \in L^2\Lambda^k(\Omega) : d\omega \in L^2\Lambda^{k+1}(\Omega)\}$$

Morally linear algebra. In finite dimensions, a self-adjoint operator T gives an orthogonal decomposition of the space into $\ker T \oplus (\ker T)^\perp$, and on $(\ker T)^\perp$ the operator is invertible.

Infinite-dimensional subtlety. For $T = \Delta$ on $\Omega^k(M)$, this is not just a formal statement: you need Sobolev spaces to make Ω^k a Hilbert space, elliptic estimates to identify the closure of the range, and Rellich-type compactness to upgrade weak solutions to smooth ones.

On a compact manifold, the Hodge theorem implies that

$$\Delta\omega = \alpha$$

Has a solution if and only if α is orthogonal to all harmonic forms. This criterion is explicit in standard hodge notes. If $\alpha = \Delta\omega$

$$(\alpha, h) = (\Delta\omega, h) = (\omega, \Delta h) = 0$$